

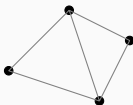
Riemannian Metric Matching

Scalable Geometric Modelling of Distributions for Anomaly Detection

The Bottleneck of Classical Diffusion Geometry

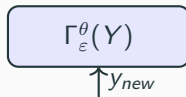
- **Classical Approach:** Relies on k -nearest neighbour (k -NN) graphs and kernel matrices to estimate the Carré du Champ (CDC) operator.
- **Scaling Limits:** Computational and memory costs scale with dataset size $\mathcal{O}(N^2)$.
- **Out-of-Sample Inference:** Processing a single new query requires recomputing pairwise distances.

k -NN Graph
Memory: $\mathcal{O}(N^2)$



Replaced by
 $\longrightarrow$

Neural Surrogate
Memory: $\mathcal{O}(1)$ inference



Conditional Metric Matching

- **Core Concept:** Train a neural network to regress the Riemannian metric tensor directly via a denoising probabilistic framework.
- **Mechanism:** Extends denoising diffusion objectives to generalise classical diffusion maps to continuous neural representations.
- **Sample-Wise Training:** Computes the conditional Carré du Champ operator over random perturbations of the data.

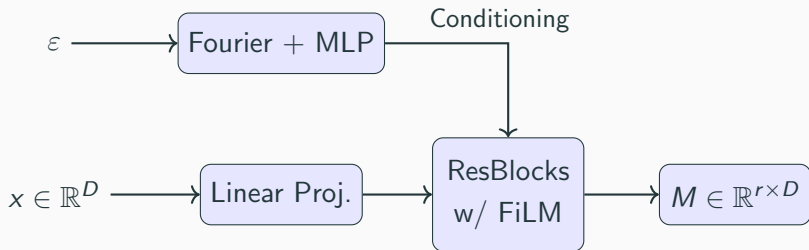
The Low-Rank Training Objective

- **Parametrisation:** The network outputs a low-rank matrix $M_{\epsilon}^{\theta}(Y) \in \mathbb{R}^{r \times D}$. The metric is formed as $\Gamma = M^T M$.
- **Rank Constraint:** To avoid topological obstructions, the rank must satisfy $r \geq 2d - 1$, where d is the intrinsic dimension.
- **Loss Function ($\mathcal{L}_{LR}$):** Avoids constructing full $D \times D$ matrices.

$$\mathcal{L}_{LR} = \mathbb{E}_{X,Y} \left[\left\| MM^T \right\|_F^2 - \frac{1}{\epsilon} \|M(X - Y)\|_2^2 \right]$$

Neural Architecture

- **Inputs:** Ambient coordinate vector $x \in \mathbb{R}^D$ and bandwidth scalar ε .
- **Bandwidth Conditioning:** ε is encoded via Fourier features to resolve fine-scale geometric structures.
- **Modulation:** Noise embedding is injected into residual blocks via Feature-wise Linear Modulation (FiLM).



Application: Time-Series Anomaly Detection

- **Phase Space:** Time series mapped to $\mathbb{R}^L$.
- **Tangent Space Extraction:** SVD on the predicted metric $M = U\Sigma V^T$.

$$\Gamma = (U\Sigma V^T)^T (U\Sigma V^T) = V\Sigma U^T U\Sigma V^T = V\Sigma^2 V^T$$

The matrix $P = VV^T$ forms the orthogonal projector onto $T_x\mathcal{M}$. Truncated to the top d intrinsic dimensions.

- **Orthogonal Directional Error:** Measures the deviation of temporal transitions Δx from the learned continuous geometry.

$$\text{Anomaly Score} = \|\Delta x - P\Delta x\|_2^2$$